

§ 0.4

$$\begin{cases} \sin \theta \\ \cos \theta \\ \tan \theta \end{cases}$$

$$\begin{cases} \sec \theta = \frac{1}{\cos \theta} \\ \csc \theta = \frac{1}{\sin \theta} \\ \cot \theta = \frac{1}{\tan \theta} \end{cases}$$

$$\textcircled{1} \text{ radian} = \text{度 (degree)} \times \frac{\pi}{180^\circ}$$

$$\textcircled{2} \text{ degree} = \text{radian} \times \frac{180^\circ}{\pi}$$

In exercises 1 and 2, convert the given radians measure to degrees.

1. ~~(a)~~ $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{4\pi}{3}$

2. ~~(a)~~ $\frac{3\pi}{5}$ (b) $\frac{\pi}{7}$ (c) 2 (d) 3

In exercises 3 and 4, convert the given degrees measure to radians.

3. (a) 180° (b) 270° (c) 120° (d) 30°

~~4.~~ (a) 40° (b) 80° (c) 450° (d) 390°

1. (a) $\frac{\pi}{4} \times \frac{180^\circ}{\pi} = 45^\circ$

(b) $\frac{\pi}{3} \times \frac{180^\circ}{\pi} = 60^\circ$

2. (a) $\frac{3\pi}{5} \times \frac{180^\circ}{\pi} = 108^\circ$

(b) $\frac{\pi}{7} \times \frac{180^\circ}{\pi} = \frac{108^\circ}{7}$

4. (a) $40^\circ \times \frac{\pi}{180^\circ} = \frac{2}{9}\pi$

(b) $80^\circ \times \frac{\pi}{180^\circ} = \frac{4\pi}{9}$

(c) $450^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{2}$

(d) $390^\circ \times \frac{\pi}{180^\circ} = \frac{13\pi}{6}$

In exercises 5–14, find all solutions of the given equation.

5. $2 \cos x - 1 = 0$

6. $2 \sin x + 1 = 0$

7. $\sqrt{2} \cos x - 1 = 0$

8. $2 \sin x - \sqrt{3} = 0$

9. $\sin^2 x - 4 \sin x + 3 = 0$

10. $\sin^2 x - 2 \sin x - 3 = 0$

11. $\sin^2 x + \cos x - 1 = 0$

12. $\sin 2x - \cos x = 0$

13. $\cos^2 x + \cos x = 0$

14. $\sin^2 x - \sin x = 0$

5. $2 \cos x - 1 = 0 \Rightarrow 2 \cos x = 1$

$$\Rightarrow \cos x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{3} + 2k\pi, k \in \mathbb{Z} \text{ or } x = \frac{5\pi}{3} + 2k\pi, k \in \mathbb{Z}$$

9. $\sin^2 x - 4 \sin x + 3 = 0 \Rightarrow (\sin x - 1)(\sin x - 3) = 0$

$$\Rightarrow \sin x = 1 \text{ or } \sin x = 3$$

$$\Rightarrow x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

10. $\sin^2 x - 2 \sin x - 3 = 0 \Rightarrow (\sin x - 3)(\sin x + 1) = 0$

$$\Rightarrow \sin x = -1$$

$$\Rightarrow x = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

11. $\sin^2 x + \cos x - 1 \Rightarrow (1 - \cos^2 x) + \cos x - 1 = 0$

$$\Rightarrow \cos x(-\cos x + 1) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \cos x = 1$$

$$\Rightarrow x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \text{ or } x = 2k\pi, k \in \mathbb{Z}$$

$$12. \sin 2x - \cos x = 0 \Rightarrow 2\sin x \cos x - \cos x = 0$$

$$\Rightarrow \cos x (2\sin x - 1) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \text{ or}$$

$$x = \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z} \text{ or}$$

$$x = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

* Review + Push progress (Preview)

量 $\hat{v}$

$$1. a \sin x + b \cos x = \sqrt{a^2 + b^2} \left(\sin x \times \frac{\overset{\cos \theta}{a}}{\sqrt{a^2 + b^2}} + \cos x \times \frac{\overset{\sin \theta}{b}}{\sqrt{a^2 + b^2}} \right)$$
$$= \sqrt{a^2 + b^2} \sin(x + \theta), \text{ where } \begin{cases} \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \\ \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \end{cases}$$

$$2. a \cos x - b \sin x = \sqrt{a^2 + b^2} \left(\cos x \cdot \frac{a}{\sqrt{a^2 + b^2}} - \sin x \cdot \frac{b}{\sqrt{a^2 + b^2}} \right)$$
$$= \sqrt{a^2 + b^2} \cos(x + \theta), \text{ where } \begin{cases} \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \\ \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \end{cases}$$

§ 0.5

$f(x)$ 的 domain = x 所有可在 $f(x)$ 取值的 set

e.g. $f(x) = \frac{1}{x}$

Dom: $x \neq 0$ or $(-\infty, 0) \cup (0, \infty)$

e.g. $f(x) = \sqrt{x}$

Dom: $x \geq 0$ or $[0, \infty)$

Functions 的平移和伸缩

1. 鉛直平移: if $c > 0$,

$$\begin{cases} f(x) + c : \text{向上 } c \text{ units} \\ f(x) - c : \text{向 } \textcolor{red}{F} \text{ } c \text{ units} \end{cases}$$

2. 水平平移:

$$\begin{cases} f(x - c) : \text{向 } \textcolor{red}{右} \text{ } c \text{ units} \\ f(x + c) : \text{向左 } c \text{ units} \end{cases}$$

3. 鉛直伸縮:

$$cf(x) : f(x) \text{ 鉛直伸縮 } c \text{ 倍}$$

4. 水平伸縮:

$$f(cx) : f(x) \text{ 水平伸縮 } \frac{1}{c} \text{ 倍}$$

37. Prove that, for some constant β ,

$$4 \cos x - 3 \sin x = 5 \cos (x + \beta).$$

Then, estimate the value of β .

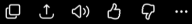
38. Prove that, for some constant β ,

$$2 \sin x + \cos x = \sqrt{5} \sin (x + \beta).$$

Then, estimate the value of β .

52. A person who is 6 feet tall stands 4 feet from the base of a light pole and casts a 2-foot-long shadow. How tall is the light pole?
53. A surveyor stands 80 feet from the base of a building and measures an angle of 50° to the top of the steeple on top of the building. The surveyor figures that the center of the steeple lies 20 feet inside the front of the structure. Find the distance from the ground to the top of the steeple.

52. 一個人身高 6 英尺，站在距離燈柱底部 4 英尺處，並投射出長 2 英尺的影子。請問這根燈柱有多高？



交流电路

55. In an AC circuit, the voltage is given by $v(t) = v_p \sin(2\pi ft)$, where v_p is the peak voltage and f is the frequency in Hz. A voltmeter actually measures an average (called the **root-mean-square**) voltage, equal to $v_p / \sqrt{2}$. If the voltage has amplitude 170 and period $\pi/30$, find the frequency and meter voltage.

振幅

$$\textcircled{1} v(t) = V_p \sin(2\pi ft)$$

$$\textcircled{2} V_{\text{rms}} = \frac{V_p}{\sqrt{2}}$$

$$f = ?, V_{\text{rms}} = ?$$

